

# OPTIMISING BUILDING ENERGY USE

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Toward the COP31

## 25% TARGET

A data-driven  
framework to  
reduce building  
energy use  
with CBECS

# THE PROBLEM

Our ambition is clear.

# 25%

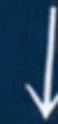
reduction in building  
energy use toward COP31

CBECS contains many variables,  
but not all are directly upgradable.

Raw data alone doesn't deliver savings.

We need to identify **what can  
actually be changed.**

RAW CBECS  
DATA



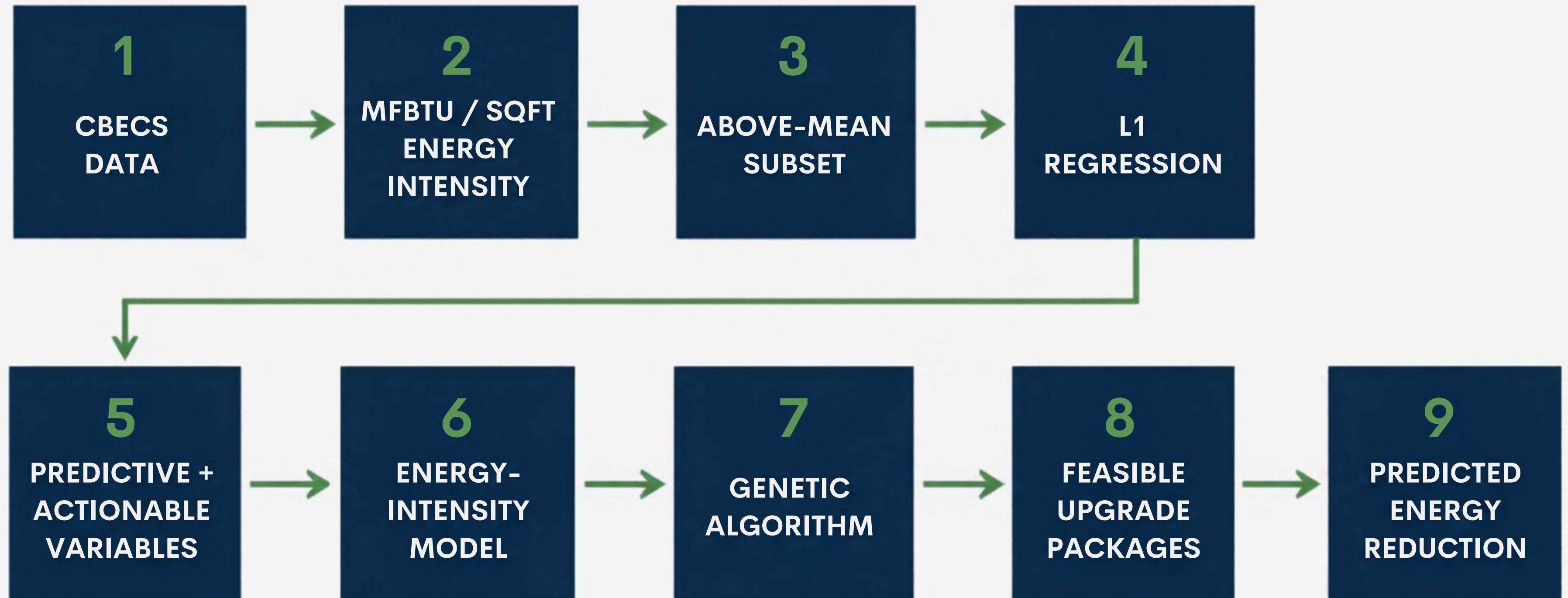
FILTER &  
ANALYZE



**ACTIONABLE  
RETROFIT  
DECISIONS**

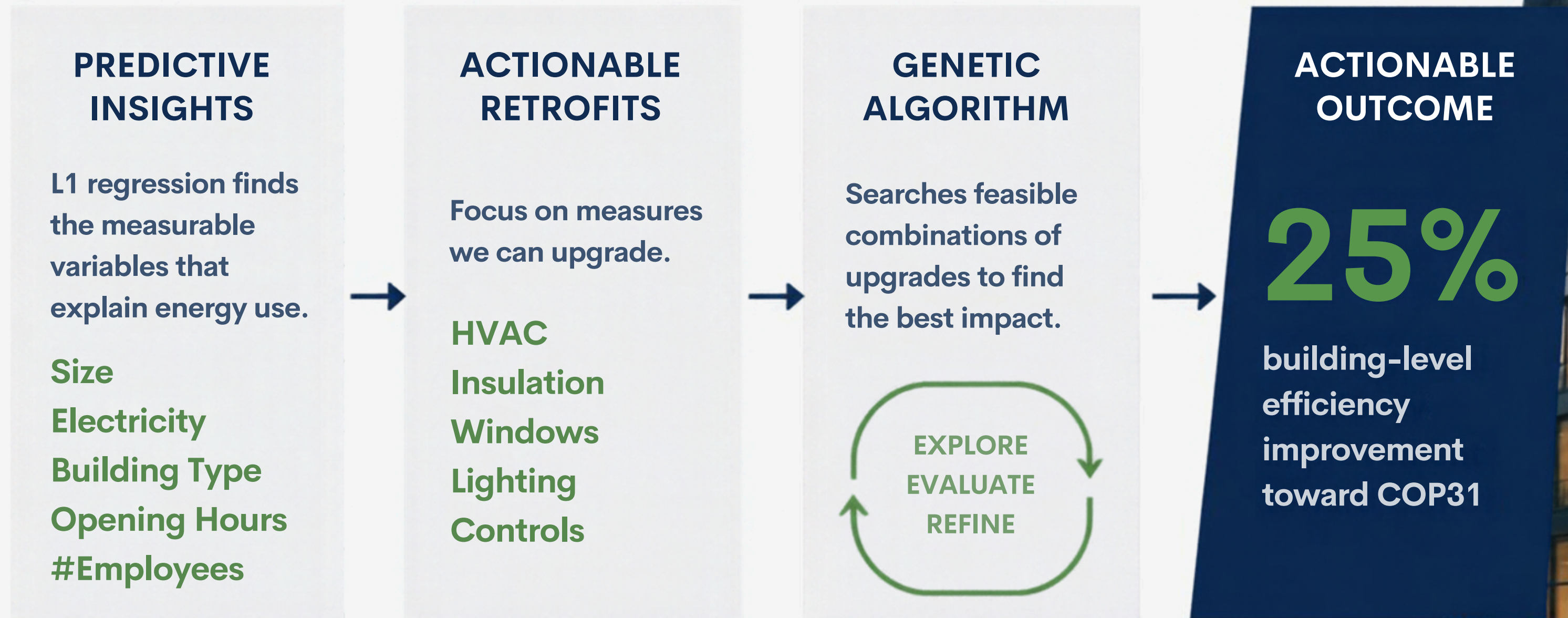


# OUR END-TO-END METHOD



# TURNING PREDICTION INTO ACTION

From model insight to upgrade decisions that deliver real savings.



# WHAT WE FOUND

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Maximum predicted efficiency improvement

8.8%

Below the 25% target,  
but a meaningful proof of concept



# 8.8% IS NOT THE CEILING

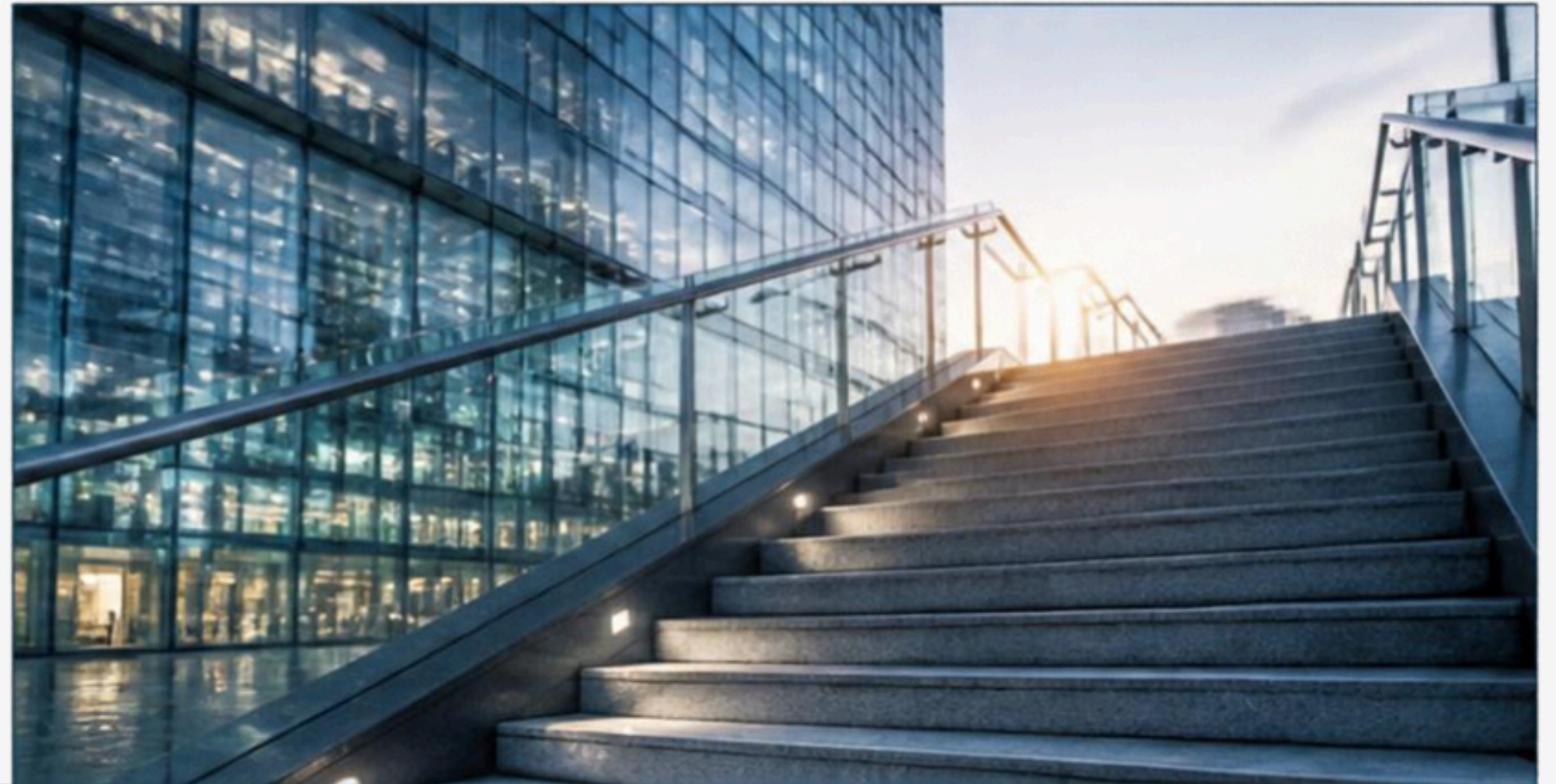
Limitations today. A credible path to 25%.

## WHAT LIMITED US

Historical data limits representation of current technologies

Time constraints limited GA tuning and sample size

Variable quality and model fit can be improved. Linear regression model quality  $R^2 \approx 0.47$

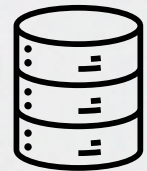


BETTER DATA → STRONGER MODELLING → BROADER OPTIMISATION

**A CREDIBLE PATH TOWARD 25%**

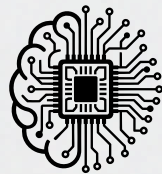
# ACKNOWLEDGEMENT

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## **DATA**

U.S. ENERGY INFORMATION ADMINISTRATION (EIA), 2018 CBECS.



## **AI DISCLOSURE**

CHATGPT ASSISTED WITH CODING, DEBUGGING, ANALYSIS WORKFLOWS, AND PRESENTATION PREPARATION. FINAL RESULTS AND CONCLUSIONS WERE REVIEWED BY THE PROJECT TEAM.



## **RESEARCH DISCLAIMER**

RESULTS ARE MODEL-BASED ESTIMATES, NOT GUARANTEED OR MEASURED ENERGY SAVINGS.

## **ACKNOWLEDGEMENTS**

OUR MENTORS, CHALLENGE ORGANIZERS, AND CONTRIBUTORS.